Meshtastic Custom App Integration Guide

Complete Technical Specification, Protocol Buffers, and AI Migration Blueprint

BLE GATT API | Protobuf Encoding | Hardware: Seeed XIAO ESP32C6

> **Hardware Compatibility Profile:** Configured for your custom node (Node ID: `0xa3999448`, BLE Name: `Meshtastic_9448`, Default Bluetooth PIN: `123456`). This guide details all steps and provides a complete prompt to adapt any custom mobile (Flutter/React Native/iOS/Android), web, or desktop application.

1. BLE RADIO SPECIFICATION & GATT STRUCTURE

GATT Attribute	UUID	Property	Function in Custom App
Meshtastic Service	`6ba1b218-15a8-461f-9fa8-5dcae273eafd`	Primary Service	Filter UUID for BLE scanner to discover the node.
`ToRadio`	`f75c76d2-129e-11ed-861d-0242ac120002`	Write Without Response	App sends serialized `ToRadio` Protobuf packets to node.
`FromRadio`	`2c55e69e-4993-11ed-b878-0242ac120002`	Read / Notify	Node sends serialized `FromRadio` Protobuf packets to app.
`FromNum`	`ed9da18c-a800-4f66-a670-aa7547e34453`	Notify	Sequence counter notifying app when a packet is ready to read.

2. IMPLEMENTATION STEPS FOR YOUR APP

Step 1: BLE Scanning & Fixed-PIN Pairing

- Scan specifically for devices advertising Service UUID `6ba1b218-15a8-461f-9fa8-5dcae273eafd` or names prefixed with `Meshtastic_`.
- When pairing, intercept the BLE PIN request and supply the headless default fixed passkey: `123456`.

Step 2: Connection Handshake & Configuration Sync

- Enable notifications on the `FromRadio` characteristic.
- Send a `ToRadio` protobuf with `want_config_id` set to request current node configuration, channel settings, and mesh node database.
- The node responds with `MyNodeInfo`, `NodeInfo` list, and `RadioConfig`.

Step 3: Packet Parsing & Dispatcher (PortNum Routing)

Every message received from `FromRadio` is a `FromRadio` protobuf containing a `MeshPacket`. Check `decoded.portnum`:

Port Number Enum	Underlying Hardware	Payload Structure & Custom App Action
`PortNum.POSITION_APP`	u-blox NEO-6M GPS (D7)	Contains `Position` protobuf. Divide `latitude_i` & `longitude_i` by `1e7` to plot coordinates on your app's live map.
`PortNum.TEXT_MESSAGE_APP`	Tactile SOS Button (D6)	UTF-8 text string. Parse messages containing `"SOS"` or `"EMERGENCY"` to trigger emergency UI and victim alerts.
`PortNum.TELEMETRY_APP`	MPU6050 Accelerometer (D4/D5)	Contains `Telemetry` protobuf with acceleration, orientation, and battery data. Use for fall detection algorithms.
`PortNum.EXTERNAL_NOTIFICATION_APP`	Piezo Buzzer & LED (D0)	Send this packet to the node to remotely activate the acoustic SAR beacon (buzzer) and visual alert.

Step 4: Transmitting Commands & Search-and-Rescue Actions

- **Trigger Audible Beacon:** Create a `MeshPacket` addressed to the target node number (or broadcast `0xFFFFFFFF`) with portnum `EXTERNAL_NOTIFICATION_APP`.
- **Send Chat / Waypoints:** Encapsulate text messages or custom SAR search grids in `TEXT_MESSAGE_APP` or custom binary payloads.

3. MASTER PROMPT FOR AI CODING AGENTS

How to Use This Prompt: Copy and paste the entire block below into your AI coding assistant (Cursor, Roo Code, Claude, Copilot, ChatGPT, or Antigravity) along with your existing codebase. It gives the AI complete context, exact protobuf schemas, and explicit implementation instructions.

📋 COPY-PASTE AI PROMPT: CONVERT CUSTOM APP TO MESHTASTIC

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### ROLE & OBJECTIVE
You are a senior mobile and IoT systems software engineer.
Our team is integrating a custom ultra-compact Search-and-Rescue (SAR) LoRa node into our existing application.
The hardware is based on the Seeed Studio XIAO ESP32-C6 running custom Meshtastic firmware (v2.8+) with:
- IndieSemiC SX1262 LoRa module (868MHz mesh network)
- u-blox NEO-6M GPS (passive NMEA telemetry)
- MPU6050 6-axis IMU accelerometer (I2C fall detection)
- Tactile Emergency SOS Push Button (Active LOW)
- 3V Piezo Buzzer & Status LED (Active HIGH SAR audible beacon)

Your task is to refactor/extend our application codebase to add full, native Meshtastic compatibility over
Bluetooth Low Energy (BLE).

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### HARDWARE & BLE SPECIFICATIONS
1. Device Advertising Name: Filter for devices starting with "Meshtastic_" (specifically "Meshtastic_9448")
2. Default Bluetooth Pairing PIN: "123456" (Fixed headless PIN)
3. Meshtastic Service UUID: 6ba1b218-15a8-461f-9fa8-5dcae273eafd
4. Characteristics:
   - ToRadio (Write without response): f75c76d2-129e-11ed-861d-0242ac120002
   - FromRadio (Read / Notify): 2c55e69e-4993-11ed-b878-0242ac120002
   - FromNum (Notify): ed9da18c-a800-4f66-a670-aa7547e34453

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### REQUIRED ARCHITECTURAL MODULES TO IMPLEMENT

1. MeshtasticBleManager:
   - Handle scanning, auto-reconnect, and pairing with passkey "123456".
   - Subscribe to FromRadio and FromNum notifications.
   - Implement ToRadio packet framing (max 512 bytes per BLE chunk).
   - On connection, send initial handshake ToRadio packet with want_config_id: 1.

2. Protobuf Serialization / Deserialization:
   - Integrate the official Meshtastic Protobuf schemas (ToRadio, FromRadio, MeshPacket, Position, Telemetry,
Data).
   - Parse all incoming FromRadio bytes into strongly-typed MeshPacket structures.

3. Live SAR Map & GPS Tracker:
   - Intercept packets with PortNum == POSITION_APP.
   - Extract latitude = latitude_i / 1e7, longitude = longitude_i / 1e7, altitude, and timestamp.
   - Update our app's map UI in real-time with node marker locations, track history, and battery state.

4. Emergency SOS Listener:
   - Intercept packets on PortNum == TEXT_MESSAGE_APP or ALERT_APP containing "SOS" or emergency triggers.
   - Immediately raise high-priority visual banner, audible alarm, and lock camera focus onto victim coordinates.

5. Remote Audible Beacon Controller:
   - Add a button in our UI labeled "Trigger SAR Siren / Buzzer".
   - When tapped, construct and transmit a MeshPacket to the target node with PortNum == EXTERNAL_NOTIFICATION_APP.
   - This fires GPIO 0 on the ESP32-C6, sounding the 3V SMD piezo buzzer to locate victims in the field.

6. Fall & Motion Detection:
   - Intercept PortNum == TELEMETRY_APP to parse accelerometer readings from the MPU6050.
   - Flag sudden impact or extended motionless state (> 3 minutes) as a potential man-down condition.

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### OUTPUT REQUIREMENTS
1. Analyze our project structure and specify exactly which files need to be modified or created.
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2. Provide clean, production-ready, modular code adhering to our project's design patterns and state management.
3. Include error handling for BLE disconnections, packet retries, and corrupted protobuf frames.
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4. QUICK VERIFICATION CHECKLIST

Test Item	Expected Result	Validation Method
BLE Discovery	App discovers `Meshtastic_9448` within 3 seconds of scan.	Open App → Tap "Scan for Rescue Nodes".
PIN Authentication	Pairs smoothly without PIN rejection error.	App supplies `123456` when prompted.
GPS Position Ingestion	Map marker appears at GPS coordinates once satellites lock.	Inspect `POSITION_APP` packet callback.
Buzzer Activation	Piezo buzzer emits loud alarm on physical board.	Tap "Sound Siren" in custom app UI.